

RCIM Model-Bank Reproduction Open-Cell Full-Matrix Closure Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the exact-paper RCIM Model-Bank Reproduction open-cell full-matrix closure wave prepared in:

- doc/reports/campaign_plans/track_1/exact_paper/2026-04-20-23-50-13_track1_open_cell_full_matrix_closure_campaigns_plan_report.md

The batch targeted only the still-open family-target pairs in the canonical RCIM Model-Bank Reproduction progress surface:

- Table 2 - Amplitude MAE Full-Matrix Replication
- Table 3 - Amplitude RMSE Full-Matrix Replication
- Table 4 - Phase MAE Full-Matrix Replication
- Table 5 - Phase RMSE Full-Matrix Replication
- campaign name: track1_open_cell_full_matrix_closure_campaigns_2026_04_20_23_50_13
- report timestamp: 2026-04-21-14-58-00
- completed family campaigns: 9
- completed validation runs now available in canonical local review paths: 756
- failed validation runs: 0
- locally reconstructed relaunch validation artifacts: 459
- later recovered first-launch MLP validation artifacts: 297
- promoted targeted pairs: 17/28
- retained canonical baseline pairs: 11/28
- aggregate campaign artifact root:
output/training_campaigns/track1/exact_paper/forward/open_cell_full_matrix_closure/shared/track1_open_cell_full_matrix_closure_campaigns_2026_04_20_23_50_13

Operational Outcome

- the eight relaunched family retry campaign folders now expose:
 - `campaign_leaderboard.yaml`
 - `campaign_best_run.yaml`
 - `campaign_best_run.md`
- the aggregate campaign root now exposes the final 459-entry locally reconstructed leaderboard and deterministic bookkeeping representative for the relaunched non-MLP family batch
- `doc/running/active_training_campaign.yaml` now records the canonical results report path for the completed open-cell closure wave
- the first MLP launch artifact gap is now closed locally:
 - all 297 first-launch validation folders are present in the canonical validation-check output root;
 - all 297 first-launch validation reports are present in the canonical validation-report root
- the recovered MLP first-launch retry evidence does not beat the already accepted baseline representative or change the canonical Table 2-5 envelope; MLP therefore remains pinned to the accepted baseline representative in the family summary
- the canonical benchmark now reads from the better value between:
 - the already accepted benchmark baseline;
 - the new open-cell retry wave.

Family Representative Outcome

The table below uses one stable representative entry per family across the accepted baseline plus the new retry wave. For MLP, the representative remains the accepted baseline entry because the fully recovered first-launch retry evidence still does not beat that baseline representative.

Family	Best Run	Scope	Closure Score	Met	Near	Open
MLP	<code>baseline_mlp_phase_3</code>	<code>phases_only</code>	0.750	1	1	0
RF	<code>track1_rf_amplitude_240_closure_attempt_07</code>	<code>amplitudes_only</code>	1.000	2	0	0
DT	<code>track1_dt_phase_162_closure_attempt_03</code>	<code>phases_only</code>	1.000	2	0	0
ET	<code>track1_et_phase_240_closure_attempt_03</code>	<code>phases_only</code>	1.000	2	0	0
ERT	<code>track1_ert_phase_162_closure_attempt_12</code>	<code>phases_only</code>	1.000	2	0	0
GBM	<code>track1_gbm_phase_162_closure_attempt_12</code>	<code>phases_only</code>	1.000	2	0	0
HGBM	<code>track1_hgbm_phase_162_closure_attempt_07</code>	<code>phases_only</code>	1.000	2	0	0
XGBM	<code>track1_xgbm_amplitude_81_closure_attempt_08</code>	<code>amplitudes_only</code>	1.000	2	0	0
LGBM	<code>track1_lgbm_phase_162_closure_attempt_07</code>	<code>phases_only</code>	1.000	2	0	0

Aggregate Winner

- Run: track1_dt_phase_162_closure_attempt_03
- Family: DT
- Scope: phases_only
- Closure Score: 1.000
- Met / Near / Open: 2 / 0 / 0

Canonical Benchmark Outcome

Surface	Harmonics Met	Total Harmonics	Open Harmonics
Table 2 Amplitude MAE	7	10	39, 156, 162
Table 3 Amplitude RMSE	10	10	none
Table 4 Phase MAE	9	9	none
Table 5 Phase RMSE	9	9	none

RCIM Model-Bank Reproduction Closure Reading

- RCIM Model-Bank Reproduction is evaluated only from the canonical Table 2-5 full-matrix surfaces plus the accepted 10 x 19 family-bank rule.
- Harmonic-wise Table 6 evidence is postponed into RCIM Harmonic-Wise Follow-Up and is no longer part of the primary closure gate.
- current remaining non-green cells across the canonical surface: 3
- RCIM Model-Bank Reproduction remains open because at least one canonical full-matrix cell is still non-green.

Resulting Canonical State

- supporting benchmark report path: doc/reports/analysis/rcim_paper_reference/RCIM Paper Reference Benchmark.md
- supporting master summary path: doc/reports/analysis/Training Results Master Summary.md
- final closeout report path: doc/reports/campaign_results/track_1/exact_paper/forward/2026-04-21-14-58-00_track1_open_cell_full_matrix_closure_campaign_results_report.md

Final Interpretation

- This batch is operationally complete and closes the intended overnight retry wave without regressing already accepted cells.
- The earlier MLP first-launch sync incident is now closed as a bookkeeping and artifact-recovery issue rather than as an unresolved training gap.
- The recovered MLP artifact set improves family-level auditability but does not change the canonical Tables 2-5 outcome or the accepted family representative outcome.
- The closure rule now stays fully aligned with the repository-wide decision to keep RCIM Model-Bank Reproduction focused only on the four full-matrix replication tables.
- Any remaining work should therefore target only the still-open cells in Tables 2-5 , not the postponed harmonic-wise branch.